AKR-TMS

نظام مراقبة المعاملات ومخاطر العملاء

An AML and KYC system designed for Iraqi banks

A plan for transaction monitoring, sanctions screening and customer risk that meets the Central Bank of Iraq's rules, gives foreign correspondents the evidence they ask for, and never touches the core banking system.



For bank leadership and compliance heads

October 2026

1. Summary

19 June 2026
Iraq returns to the FATF grey list, with an eight-point action plan

32 banks
on the CBI's February 2026 lists: 28 barred from dollar dealings, 4 blacklisted

About 6
suspicious transaction reports per bank per year (2024)

86+
CBI scenarios every bank must run as a minimum (2025 controls)

Iraqi banks must now prove their anti-money laundering controls to people outside the bank: the Central Bank of Iraq, the FATF's reviewers and, above all, the foreign correspondent banks that decide whether an Iraqi bank can move dollars at all. Having a monitoring system is no longer enough. The system has to find what matters in Iraqi data, and show that it did.

AKR-TMS is an anti-money laundering (AML) and know-your-customer (KYC) system designed for Iraqi banks. It monitors transactions, screens customers and payments against sanctions lists, rates customer risk and runs KYC reviews, supports investigations, prepares reports for the AML/CFT Office's goAML system, and keeps the evidence of every step. It reads the bank's core banking database through a read-only connection and never changes it.

What makes it different:

- **Made for Iraq.** The CBI's full scenario catalogue, Iraq's national list and Arabic name matching come built in, not as add-ons to a Western product.
- **Read-only, and complete.** No writes to the core banking system, and a daily reconciliation that proves every record was seen.
- **Evidence by default.** Packs for the CBI, correspondents, validators and the board come from live data in minutes.
- **Explainable, with people in charge.** Rules, models and the AI assistant each have their own controls; a named person makes every decision.
- **One product for every Iraqi bank.** Conventional and Islamic, private and state-owned, with regime packs for the foreign rules each bank's correspondents apply.

The plan runs in phases: six to eight weeks of discovery and data mapping, a core system live at the first bank in about nine months after a parallel run beside its current system, then checks on payments before release and validated machine-learning models over the following six months. Further banks follow in three to five months each.

2. Why now: the position Iraqi banks are in

Back on the grey list

On 19 June 2026 the Financial Action Task Force (FATF) put Iraq back on its list of jurisdictions under increased monitoring, the “grey list” that Iraq left in 2018. Iraq’s action plan has eight items. Four of them land directly on the monitoring systems of Iraqi banks:

- more suspicious transaction reports, and better ones;
- detecting informal money transfer (hawala) activity that runs through bank accounts;
- stronger oversight of who really owns and controls companies;
- stopping the evasion of sanctions on weapons proliferation.

Dollar access now depends on each bank’s own controls

Iraq’s oil revenue is held at the Federal Reserve Bank of New York, so the United States decides whether Iraqi banks can obtain and move dollars. Since 2023 Iraqi banks have been cut off from dollar transactions in waves: 14 banks in July 2023, 8 in January 2024 and 5 more reported in February 2025. In 2024 the US Treasury went further and designated one Iraqi bank a “primary money laundering concern”, which closes the US financial system to it entirely. The CBI’s own lists of February 2026 name 28 banks that may not deal in dollars, directly or indirectly, and 4 that are blacklisted.

The way back now runs through compliance. In July 2026 the CBI announced an understanding with the US Treasury: banks that meet its compliance and governance standards and complete the first phase of its reform and re-licensing programme return first to correspondent banking in other currencies, and later regain dollar dealings. Seven banks qualified for the first step. A bank that wants to be on that path has to show its controls work.

At the end of 2024 the Central Bank of Iraq (CBI) closed the electronic platform through which it had vetted dollar transfers. Since then each bank deals directly with its own foreign correspondent banks, in Jordan, the UAE, Kuwait, Europe and the United States. Each correspondent applies its own country’s rules to its Iraqi customers, and each can end the relationship.

July 2023	US bars 14 Iraqi banks from dollar transactions
January 2024	8 more banks barred after Iran sanctions concerns
July 2024	US Treasury final rule against an Iraqi bank as a “primary money laundering concern”
End of 2024	CBI dollar platform closes; banks must deal with correspondents directly
January 2025	CBI issues detailed AML supervisory controls with explicit system requirements
February 2025	5 more banks barred, most of them Islamic banks
September 2025	UN sanctions on Iran restored; US designates four more Iraqi armed groups as terrorist organisations
February 2026	CBI lists 28 banks barred from dollar dealings and 4 blacklisted
June 2026	FATF grey-lists Iraq
July 2026	CBI and US Treasury agree a way back for banks that pass the CBI’s reform programme

- **August 2026** — CBI circulars on behavioural indicators and politically exposed persons
- **September 2026** — Deadline for each private bank's external AML assessment under the CBI's reform
- **July 2027** — New EU anti-money laundering regulation applies to every EU correspondent

The bar has been raised at home

The CBI's supervisory controls of January 2025 are unusually specific about what an AML system must do. They require real-time monitoring of account activity against what the customer declared; the CBI's own catalogue of more than 86 scenarios as a minimum; artificial intelligence and machine learning to find suspicious activity; holding a transaction that matches a scenario until the money laundering reporting officer (MLRO) clears it; screening against the UN, Iraqi, US, EU and UK lists, refreshed at least every 12 hours, with search in Arabic and English; a liaison officer in every branch with a window into the system; and AML tools verified by a CBI-approved firm. The August 2026 circulars added daily news screening of politically exposed persons and their relatives to the second degree.

The CBI's banking reform, in force since September 2025, makes that verification a test of survival. Every private bank that chose to continue had to submit, by 30 September 2026, an assessment of its anti-money laundering controls by an independent external auditor approved by the CBI. Gaps must be fixed and the file resubmitted within six months; a bank that cannot fix them may be moved to the merger or liquidation path. The auditor tests the system itself: its rule logic and alert thresholds, user permissions, the integrity of its audit trail, its links between customer checks, monitoring and screening, and whether the data behind each report can be traced.

Execution is the gap

The 2024 mutual evaluation of Iraq found that most of the rules are in place but that they do not yet work well enough: it rated the effectiveness of banks' preventive measures "moderate" and of proliferation sanctions "low". Every Iraqi bank already runs an automated monitoring system with the CBI's standard scenarios, yet none of the banks interviewed had written a single scenario of its own, and large numbers of alerts were archived every day against very few reports. Bank reports to the AML/CFT Office rose from 82 in 2022 to 672 in 2023 and 451 in 2024; that is still about six reports per bank per year. The quality is there (the Office graded 90 percent of bank reports "good" in 2022). The volume, the coverage and the evidence are not.

Correspondents have become a second supervisor

Every foreign regime reaches an Iraqi bank through its correspondents:

	WHAT IT MEANS FOR AN IRAQI BANK
United States	Dollar clearing depends on screening against US sanctions lists, including companies owned 50 percent or more by listed persons, and on coverage of Iran-related and armed-group typologies. Sanctions records must be kept for 10 years
European Union	From July 2027 every EU bank applies one regulation that requires a record of every assessment, human involvement in automated decisions, and answers to authorities within days. A new EU authority can single out an individual non-EU bank whose controls fall short; EU banks must then stop new business with it and end existing relationships unless they can show the risk is covered

	WHAT IT MEANS FOR AN IRAQI BANK
United Kingdom	Iraq counts as a high-risk country. UK banks no longer have to apply enhanced checks to every Iraqi relationship automatically, so the quality of the Iraqi bank's own controls decides how it is treated
UAE and the Gulf	UAE banks follow detailed guidance on monitoring and screening (daily re-screening, data testing, documented tuning) and expect the same from their partners. Kuwait and Lebanon are grey-listed too, so their banks pass the pressure on

The common thread is evidence. A correspondent cannot visit the bank, does not read Arabic, and will not take the bank's word for it. It wants to see which lists are screened and how fast they are updated, which scenarios run and how they were tested, how many alerts are waiting and for how long, and who approved each change.

> **What this means.** An Iraqi bank's AML system is no longer a back-office tool. It decides whether the bank can do business in dollars. It has to do three things at once: meet the CBI's explicit requirements; find what matters in Iraqi data, which is largely in Arabic; and prove all of it to people outside the bank.

3. What the rules require, and how they converge

Iraq first

Iraqi law and the CBI's instructions always come first. The research behind this plan read the AML/CFT Law No. 39 of 2015, the CDD instructions of 2023, the CBI supervisory controls of 2025 and the CBI circulars of August 2026 in full. Their requirements for an AML system, condensed:

AREA	WHAT THE CBI REQUIRES
Monitoring	Real-time monitoring of all account activity against declared activity; the full CBI scenario catalogue as a minimum, plus the bank's own scenarios; AI and machine-learning analytics
Holding transactions	A transaction matching a scenario is held until the MLRO clears it
Customer risk	High, medium and low ratings (up to five levels) on customer, product, channel and geography; re-rating on events; review at least every two years; KYC refresh at least yearly
Screening	UN, Iraqi national list, US, EU and UK lists, with World-Check named as the data source; all parties including beneficial owners; refresh at least every 12 hours; Arabic and English search
PEPs	Relatives to the second degree and close associates; daily local and regional news screening; status kept after leaving office
Wire transfers	Complete originator and beneficiary data including purpose; incomplete incoming transfers detected and rejected if not fixed
Branches	A liaison officer in every branch with a branch window in the AML system; monthly report by the 10th
Reporting	STRs through goAML, including attempted transactions and any amount, with reasons; written decisions, also when the MLRO decides not to report
Records	At least five years after the relationship ends, and seven years of AML data for the reform's external assessment; transactions can be reconstructed; off-site backup

AREA	WHAT THE CBI REQUIRES
Assurance	AML tooling verified by a CBI-approved firm; three lines of defence; reports to the board and audit committee

International standards

The FATF's 40 Recommendations (revised to October 2025) set the global baseline: customer due diligence, beneficial ownership, politically exposed persons, payment transparency, suspicious transaction reporting and the freezing of terrorist and proliferation funds "without delay", which the FATF defines as within hours. The UN Security Council's consolidated list, including the Iran sanctions restored in September 2025, applies to every Iraqi bank. The Basel Committee, the Wolfsberg Group of global banks and the Egmont Group of financial intelligence units add detailed guidance on monitoring, screening and reporting that correspondents use as their checklist.

Foreign rules converge on one specification

US, EU, UK and UAE rules were written separately and years apart, yet they now describe almost the same screening and monitoring system: complete and current lists checked against the official source; every party and identifier screened, in the original script and in transliteration; ownership and control taken into account; matching tested before and after every change; re-screening on every list change; backlogs under control; and every decision recorded and explainable. AKR-TMS is built to the union of these rules, so that one system satisfies the CBI and every correspondent at once, and each bank switches on the foreign "regime packs" that apply to it.

> **The research behind this plan** reviewed Iraqi law and CBI instructions, the FATF standards, UN sanctions, the Basel, Wolfsberg and Egmont guidance, US, EU, UK and Gulf regimes, more than 50 AML and KYC suppliers, and about 40 enforcement cases. It produced more than 1,200 specific product requirements, each traced to its source.

4. What the market offers, and where it falls short

More than 50 AML and KYC suppliers were reviewed, from the global suites and the core banking vendors to this year's AI start-ups and the regional suppliers already working in Iraq. The best of them do many things well, and AKR-TMS adopts those as its minimum. None documents the whole of what an Iraqi bank needs.

KIND OF SUPPLIER	EXAMPLES	STRONG AT	WEAK FOR AN IRAQI BANK
Global AML suites	Oracle, SAS, NICE Actimize, SymphonyAI	Breadth, rule libraries, case management	Cost, heavy data loading, gaps in Arabic and goAML
AI detection specialists	Quantexa, Hawk, Feedzai, Tookitaki	Network analysis; models that rank alerts	Mostly foreign cloud or big-data stacks; no Arabic

KIND OF SUPPLIER	EXAMPLES	STRONG AT	WEAK FOR AN IRAQI BANK
Investigation automation	Silent Eight, Napier, Lucinity, Unit21, Flagright	Faster alert review; modern case tools	Mostly cloud; more autonomy than Iraqi rules allow
Screening engines and list data	LexisNexis, Eastnets, FinScan, Swift, LSEG World-Check, Dow Jones	Sanctions and PEP lists; payment screening	Arabic and Iraqi names handled poorly; no data supplier publishes its Iraqi coverage
KYC and onboarding platforms	Fenergo, Pega, Moody's, Encompass, Uqudo	Client lifecycle workflows; identity checks	Cloud-only, and built to own the customer record with write access; only one documents the Iraqi national card
AML modules of core banking vendors	Temenos, Azentio (iMAL), Capital Banking Solutions, Oracle (FLEXCUBE)	A ready link to their own core system; one contract	End-of-day feeds; no published Arabic matching or full CBI pack. The core system most Iraqi banks run has no AML module
Regional suppliers	Pio-Tech (Jordan), Eastnets (UAE)	Iraqi or regional customers; goAML and customer-risk modules; Arabic-speaking teams	No published accuracy or prices; detection depth unclear

What Iraqi banks run today. Every Iraqi bank already runs a monitoring system (section 2), so AKR-TMS replaces or sits beside an existing tool rather than filling a gap. Banks' annual reports name their tools, often several per bank: FICO Siron for monitoring at two of the largest private banks, Firco for payment screening at two banks, core banking vendors' AML modules at four banks, and goAML, customer-risk and scenario modules from Pio-Tech, a Jordanian supplier that lists 29 banks in Iraq among its references. None of the largest global suites has an Iraqi customer. Auditors at two banks still describe the CBI's minimum as 25 scenarios; the 2025 catalogue numbers 86.

What the best systems get right

- **Rules and models together.** Rules cover known patterns and legal duties, models rank the alerts and spot unusual behaviour, and every alert shows its reasons.
- **Peer groups and tested thresholds.** Customers are compared with similar customers, thresholds are set per group, and a sample of what falls just below a threshold is reviewed.
- **The customer, not the transaction.** Alerts are gathered per customer and per network, so an analyst investigates a person or a group, not a list of payments.
- **Changes tested before they go live.** The best products replay every rule change on past data and show the difference before anyone approves it.
- **Screening that explains itself.** Scores are broken down field by field, and a decision on a false match is remembered so that nobody reviews the same pair twice.
- **Reporting as a workflow.** Draft, approve, file, acknowledge and archive, with validation before submission.
- **One product, many banks.** SAS serves 120 Nordic banks through a shared service: one standard product, with a separate layer for each bank's settings.

Where they fall short for an Iraqi bank

WHAT AN IRAQI BANK NEEDS	WHAT THE MARKET OFFERS
A system in the bank's own data centre	Most new products, and the newest AI features of the large suites, run only in foreign clouds
Arabic names matched properly, Iraqi naming included	Arabic treated as an edge case. Oracle's on-premises name matcher states that Arabic characters "will not match"; no vendor documents handling of the mother's name, tribe or Kurdish spellings
The CBI's scenario catalogue	Global vendors ship their own libraries; the systems in Iraq that carry CBI scenarios do not show which ones, or whether the set is current
Reports to the AML/CFT Office through goAML	One global suite documents a goAML module, but not for Iraq; a regional module runs at two Iraqi banks, with no published test results
Islamic contracts monitored as contracts	No vendor documents murabaha, ijara, musharaka or mudaraba
A read-only link with proof of completeness	Heavy data loading through several staging databases; completeness rarely measured
Rules that can change quickly	TD Bank added no new scenario between 2014 and 2022
A price an Iraqi bank can pay, and local support	No published prices; global suites meter licences by volume and have no office in Iraq
Continuity	NICE Actimize is being sold; FICO and BAE sold their AML products; start-ups are being bought

> **Claims and evidence.** Vendors advertise that AI clears 75 to 98 percent of alerts automatically and removes 30 to 99 percent of false positives. The only figure compiled by a regulator from a bank in production is 7 percent of alerts closed automatically, with sampling and the option to reopen (Hong Kong Monetary Authority, 2023). AKR-TMS promises only what it can show on the bank's own data before a contract is signed: a replay of twelve months of the bank's history, compared alert by alert with the system the bank already runs.

A legal check comes first

Iraq's 2022 law against normalisation with Israel bans economic dealings with Israeli companies, with severe penalties. NICE Actimize, one of the largest AML vendors, and ThetaRay, an AI detection vendor, are headquartered in Israel. Banks should take legal advice before considering either. US vendors carry a different risk: their own sanctions policies can end licences and support for a bank that becomes subject to US measures. AKR-TMS applies the same legal check to every component and data feed it uses.

Where AKR-TMS fits

AKR-TMS is built for the gap: in the bank's own data centre, read-only, Arabic first, with the CBI catalogue, goAML reporting and Islamic contracts built in, a published price, and continuity guaranteed by source-code escrow and open export of all data. Where a bank already runs a screening engine it trusts, AKR-TMS imports that engine's alerts and adds what is missing, rather than demanding a replacement.

5. Lessons from enforcement

About 40 enforcement actions against banks and payment firms were studied, most of them from the regulators' own notices. The cause was seldom a weak algorithm. Banks were fined because the system was not given all the data, because alerts were limited to fit the staff, because nobody tested whether it worked, and because the records could not be trusted.

WHAT WENT WRONG	WHERE IT HAPPENED	WHAT AKR-TMS DOES
The system did not see all the data	TD Bank left 92 percent of its transaction volume, over USD 18 trillion, unmonitored (US, 2024). Metro Bank's monitoring, fed from a copy of its core banking data, missed 60 million transactions worth GBP 51 billion (UK, 2024)	Reconciles what it received against the core banking database every day, table by table, and shows coverage on the front page
Alerts were limited to fit the staff	US Bancorp capped alerts to match its staffing; ING's system stopped monitoring some categories after three alerts a day; by 2015 HSBC UK's rules suppressed two-thirds of its alerts	No caps and no hidden filters. Workload is managed in the queue; a threshold changes only with tested evidence and approval from outside operations
Nobody tested whether it worked	Starling screened customers against 39 of 3,088 listed names for six years (UK, 2024)	Daily tests with known answers on every list and scenario; a scenario that goes silent raises an alarm
Data the bank held was never used	Standard Chartered held fax numbers and IP addresses pointing to Iran; Swedbank Latvia saw IP addresses in Crimea	Addresses, phone prefixes and login locations feed screening and customer risk
Exemption lists switched controls off	Habib Bank's "good guy" list let at least USD 250 million through unscreened; Rabobank's "verified" customers, exempt from review, grew from fewer than 10 in 2009 to over 1,000 by 2012	Exemptions are approved by two people, expire, and are re-tested whenever a list changes
Declared activity was never checked	NatWest onboarded a jewellery business expected to handle no cash and never compared that with the cash it deposited	Expected activity from KYC is compared with actual activity in every customer review and scenario
Records were edited or withheld	Metro Bank (edited minutes), US Bancorp (altered presentations), Swedbank (narrowed answers to its New York regulator; USD 50 million, July 2026)	Logs, parameter changes and board packs are generated by the system with full history, and cannot be altered
Backlogs grew with the business	Coinbase's unreviewed alerts passed 100,000 by the end of 2021 after rapid growth	Queue clocks, ageing alarms and volume forecasts on the command centre

Closest to home. On 1 September 2026 the US Treasury proposed cutting the UAE branches of Egypt's state-owned Banque Misr off from US correspondent banking. Its case rests on flows alone: 103 potential Iranian front companies that moved about USD 1.8 billion through the branches, set against their size. A bank can compute the same ratios from its own data. AKR-TMS gives each bank this view of itself before its correspondents compute it.

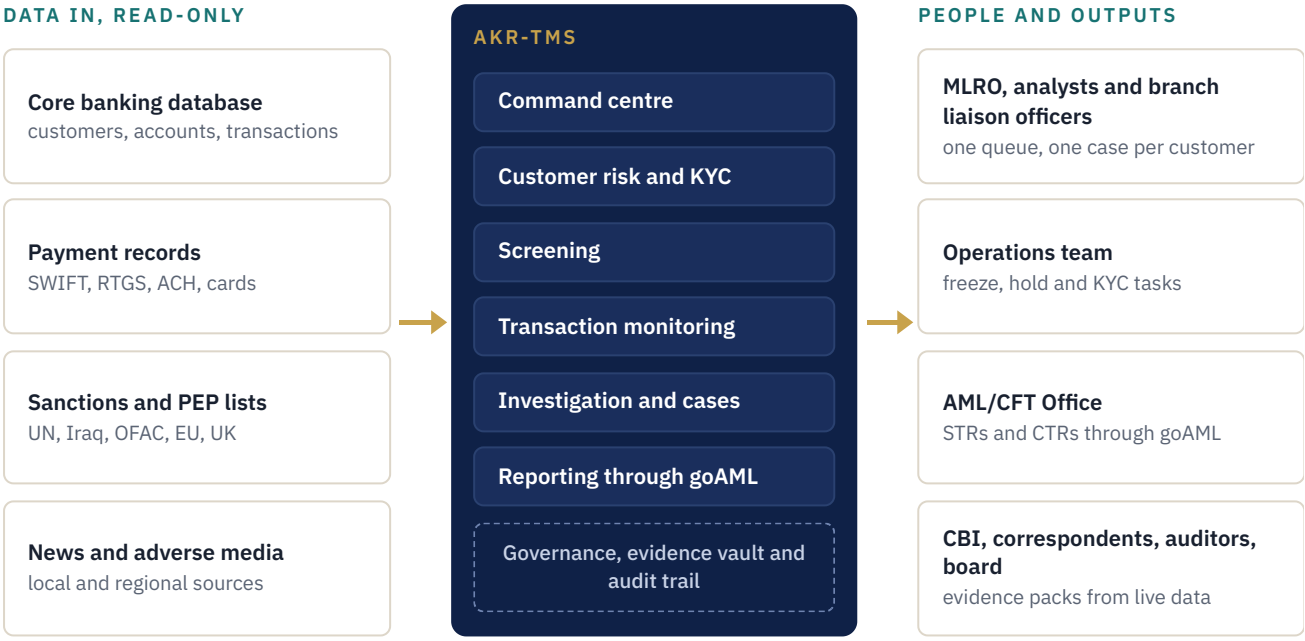
6. Design principles

Ten principles follow from the rules, from what existing systems do well and badly, and from the cases in which banks were fined. Each one shapes something a user can see in the product.

1. **Iraq first.** Built around Iraqi law, the CBI's scenario catalogue, Iraqi lists, Arabic names and dinar amounts. Foreign rules are added on top as regime packs, never the other way round.
2. **Read-only, and provably so.** AKR-TMS never writes to the core banking system and never calls its application. The bank's own database permissions guarantee it; the bank's own audit log proves it.
3. **Complete, and able to show it.** Every day AKR-TMS reconciles what it received against the core banking system, table by table, counts and totals. A scenario that cannot run because data are missing is reported, never skipped in silence.
4. **Arabic is the original.** Names are matched in Arabic script and in every common transliteration. The interface is designed right to left first, with Hijri and Gregorian dates and both styles of digits.
5. **No black boxes.** Every alert shows its reasons, every risk score its factors, every rule a plain-language description generated from its settings.
6. **People decide.** The system finds, ranks, links and drafts. A named person decides, and a second person confirms the decisions that matter.
7. **Every decision is recorded,** including the decision not to report, with its reasons. Records are kept as long as the strictest applicable rule requires, and cannot be altered.
8. **Evidence is a product.** The packs that the CBI, correspondents, validators and the board ask for come from live data in minutes, not from weeks of manual assembly.
9. **Change is governed.** Rules and models change only through testing, approval and versioning, and every change can be rolled back.
10. **Proportionate.** Effort follows risk. Low-risk customers are not buried in checks, and the system does not demand what no rule requires.

7. The system, module by module

AKR-TMS has six working modules on a shared foundation of governance, evidence and audit. All of them work from one picture of each customer, built from the core banking data, so an analyst never has to reconcile what the screening system, the monitoring system and the KYC file each say about the same person.



Command centre

The command centre is the MLRO's daily view, and the first screen a supervisor or a correspondent is shown. It answers five questions at a glance: is the data complete, are the lists current, how large and how old is the backlog, what is due to whom, and how is customer risk spread across the bank.

- Data coverage and list freshness sit at the top, because every other number depends on them.
- The backlog is shown by queue and by age, with service levels per alert type and a surge mode for the day a large list update lands.
- Every deadline is in one place: CTR batches, the branch liaison officers' monthly reports, correspondent requests, periodic reviews and annual tests.
- Each branch liaison officer gets a branch window, as the CBI requires, limited to that branch's customers and tasks.

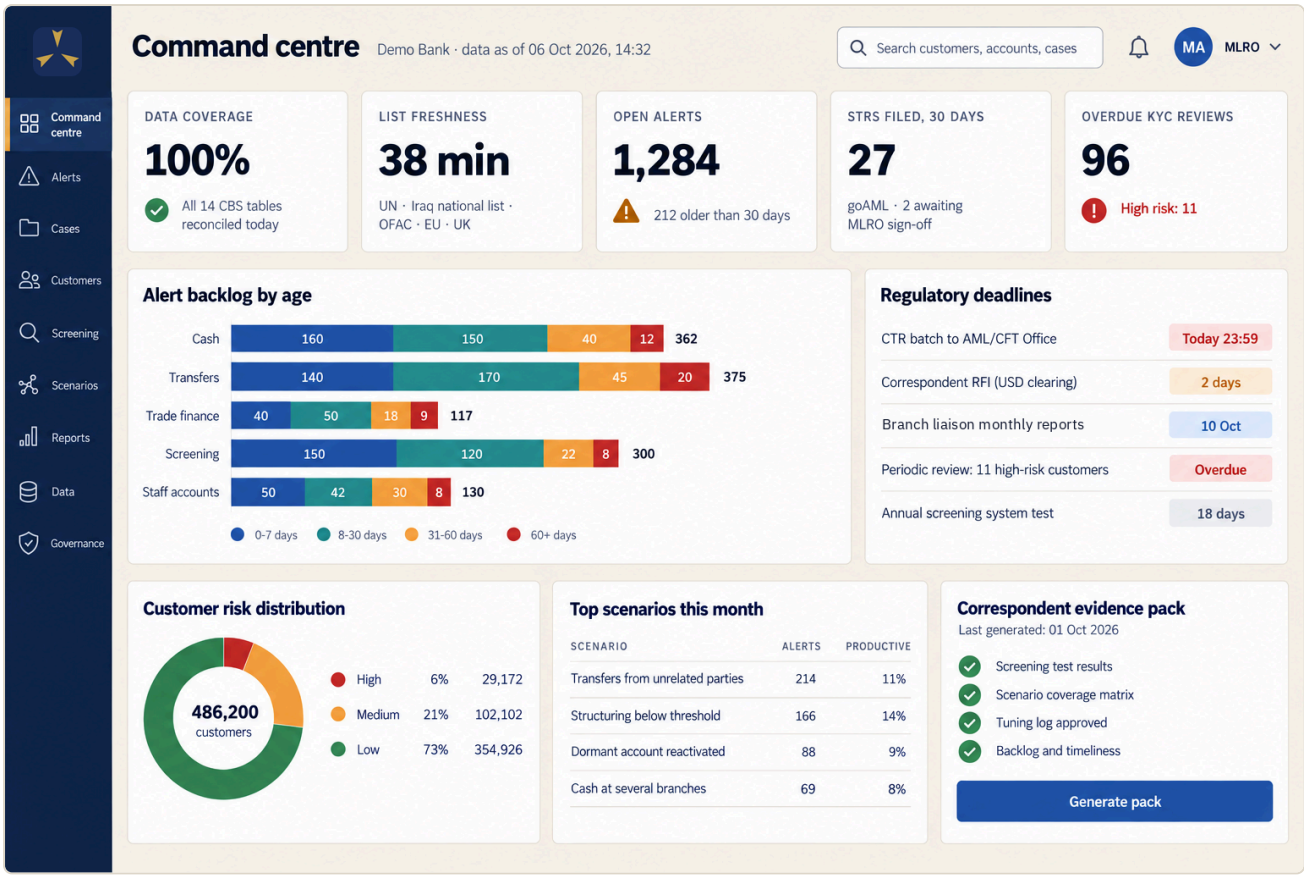


Figure 1. Command centre. Coverage, list freshness, backlog age, deadlines and risk distribution on one screen; the correspondent evidence pack is generated from the same live data. All names and figures in the mockups are fictional.

Customer risk and KYC

Customer risk is calculated, not typed in. AKR-TMS rates every customer on the four risk elements the CBI prescribes (customer, product and service, delivery channel, geography) on up to five levels, and re-rates on events instead of waiting for the next review: a list match, a sharp change in activity, an adverse news item, a new owner.

- Each score shows the factors behind it, so the rating can be explained to an auditor, a correspondent or the customer's relationship manager.
- Reviews are scheduled by risk and by the CBI's minimum intervals; overdue reviews escalate automatically.
- Enhanced due diligence is a checklist with the evidence attached: source of funds, beneficial owners, senior management approval, site visit.
- Ownership is drawn as a tree, with owners at 20 percent or more (the Iraqi threshold) highlighted.
- Documents go into an evidence vault that locks them for the retention period.
- Onboarding stays in the core banking system. AKR-TMS reads what was captured there and flags what is missing, such as an occupation, the purpose of an account or a company's owners.

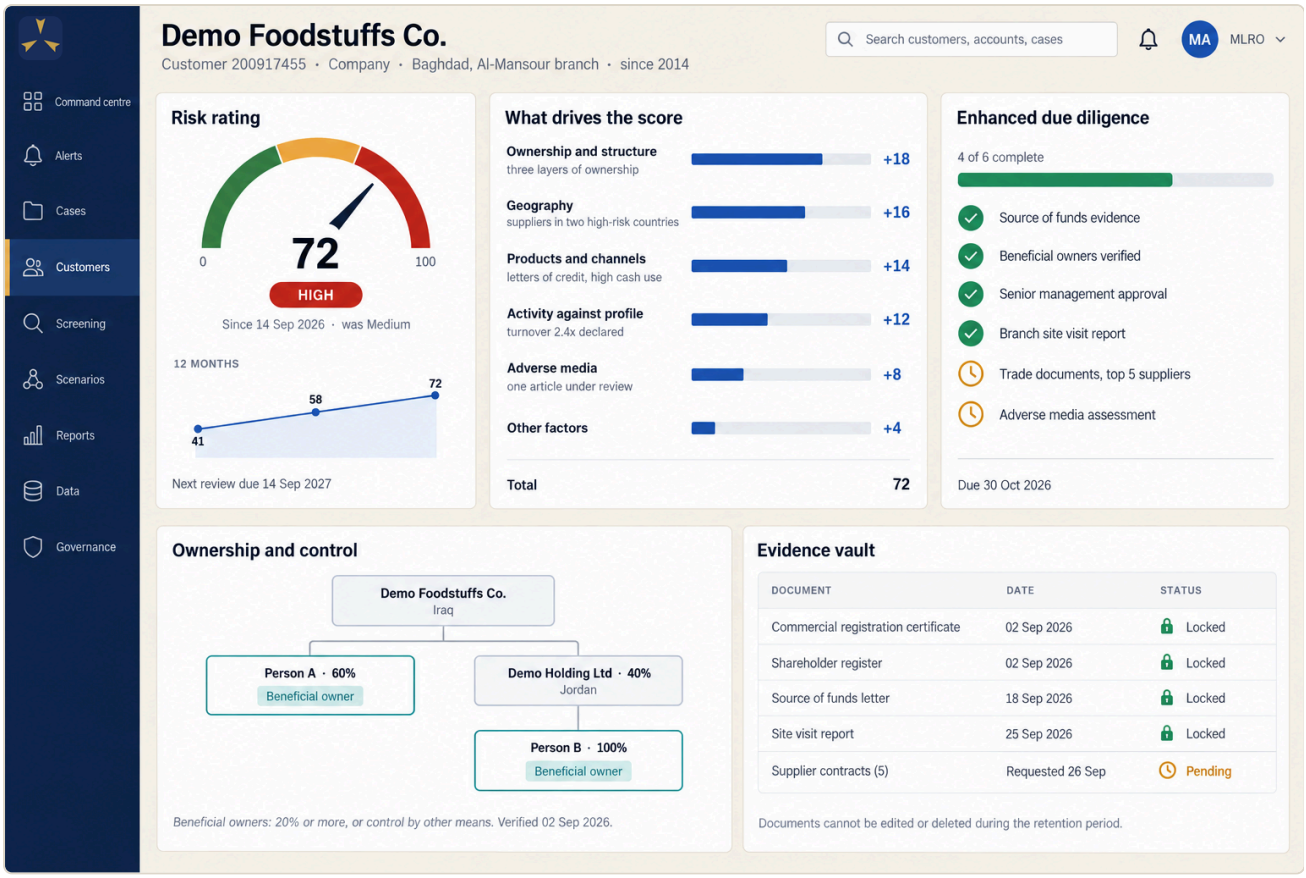


Figure 2. Customer risk profile. A score of 72 broken down into its factors, with the enhanced due diligence checklist, the ownership tree and the locked evidence beside it.

Screening

Screening is where Iraqi banks are most exposed, and where imported systems are weakest. Most names on an Iraqi bank's books are Arabic, and so are many names on the lists that matter most to it. The same name can be written in Latin letters a dozen ways (Mustafa, Moustafa, Mustapha), and a full Iraqi name chains a person's own name with the father's, the grandfather's and often a family or tribal name.

- Names are compared in Arabic script and in transliteration, name part by name part. Very common tribal names and titles such as Sayyid, Sheikh and Hajji count for less. Kurdish names are covered too. The mother's name, which Iraq's national list records for almost every entry, helps confirm or rule out a match; no engine on the market documents using it.
- The lists are the UN, Iraq's national list, OFAC (including companies owned 50 percent or more by listed persons), the EU and the UK. Each is checked against the official source on every load and refreshed well inside the CBI's 12-hour limit. Commercial data, including World-Check, which the CBI's controls name, connects through the bank's own licence.
- Every customer, beneficial owner, counterparty and payment party is re-screened whenever a list changes, not only at onboarding.
- When an analyst clears a match as the wrong person, the decision is remembered for that pair of records and reopened automatically if either record changes.
- A true match creates freeze and report tasks with their legal deadlines, such as the first report to Iraq's freezing committee within 48 hours. AKR-TMS then confirms from the core banking data

that the freeze is in place, and flags any money that leaves a frozen account afterwards, fees included. In September 2026 the UK sanctions authority fined a global bank's London branch partly for fees taken from frozen accounts.

- A built-in test harness measures matching quality before and after every change, as US, EU, UK and UAE rules now expect.

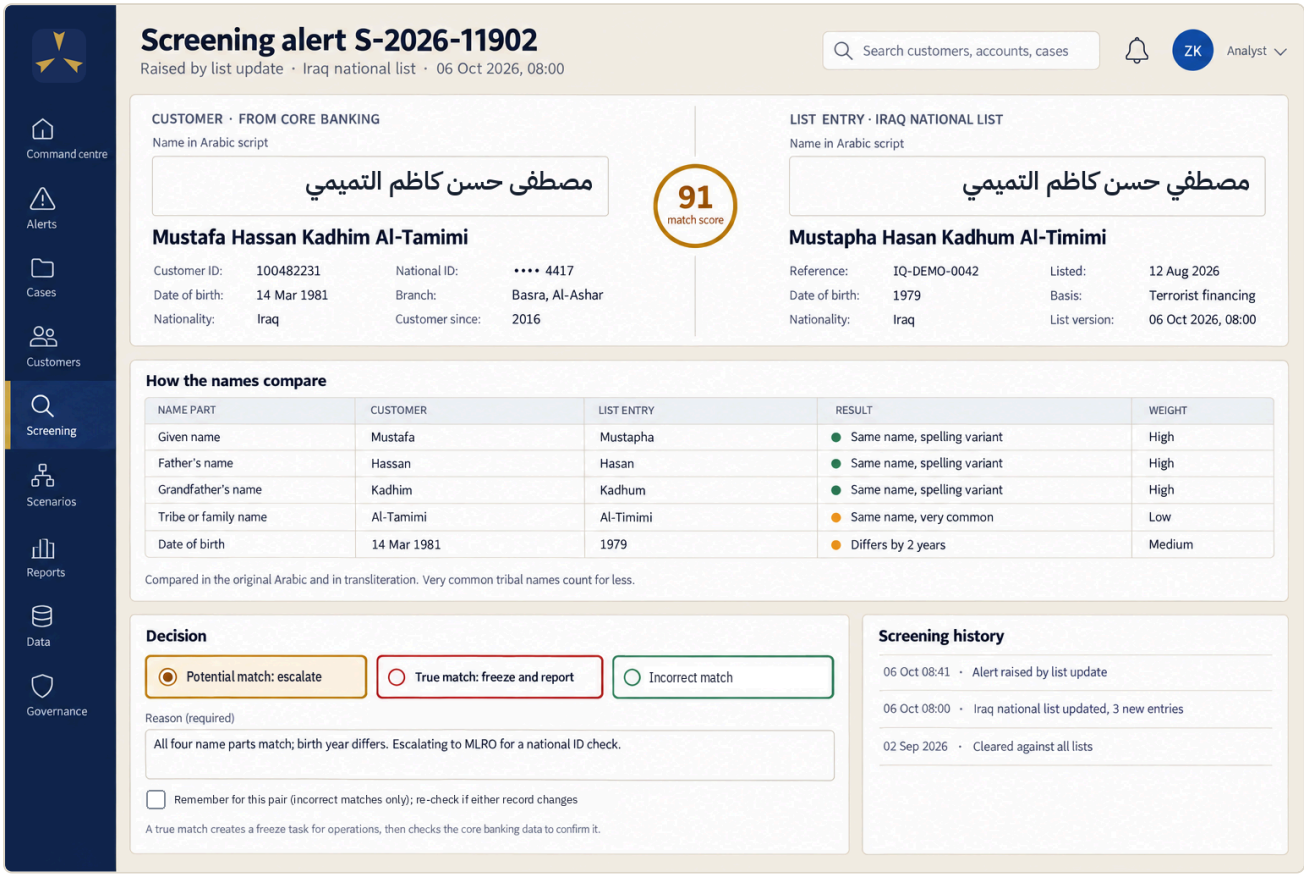


Figure 3. Screening review. The customer and the list entry are compared in Arabic script and in transliteration, name part by name part. The analyst escalates this potential match with a recorded reason; a pair is remembered only when it is cleared as the wrong person.

Transaction monitoring and the scenario studio

AKR-TMS ships with the CBI's full scenario catalogue: 86 numbered scenarios plus the product indicators, each mapped to the data it needs and to the CBI article it implements. Its library holds 161 scenarios in 18 families. Beyond the catalogue they cover the risks named in Iraq's action plan and by foreign authorities (hawala through bank accounts, sanctions evasion, front companies, trade-based laundering, misuse of dollar access), Islamic products, and public funds and payroll: corruption is the threat Iraq's national risk assessment ranks first, yet the CBI catalogue has no scenario for it. As the CBI's controls require, a bank adds such scenarios in coordination with the AML/CFT Office.

- Each scenario shows whether the bank's data can support it. Only about a quarter of the CBI's 173 scenarios and indicators can run on the core system's account, transaction and customer data alone; the rest need other modules such as cards or trade finance, payment messages, outside sources, or what branch staff observe at the counter, recorded through the branch window in AKR-TMS.

- Scenarios run per customer segment with their own thresholds, so a supermarket chain is not judged like a student.
- In the scenario studio the compliance team changes a threshold, reads a plain-language description of what the scenario will now do, and tests the change on months of past data before anyone approves it. The test includes a review of a sample of cases that would no longer alert.
- Every change goes through a change request with a second-line review and the MLRO's approval, and every version is kept.
- Alerts about the same customer are grouped into one case, so analysts investigate customers, not alerts.
- Monitoring runs continuously on data that are minutes old. Holding a payment before it leaves needs the optional phase-2 connection described in section 8.

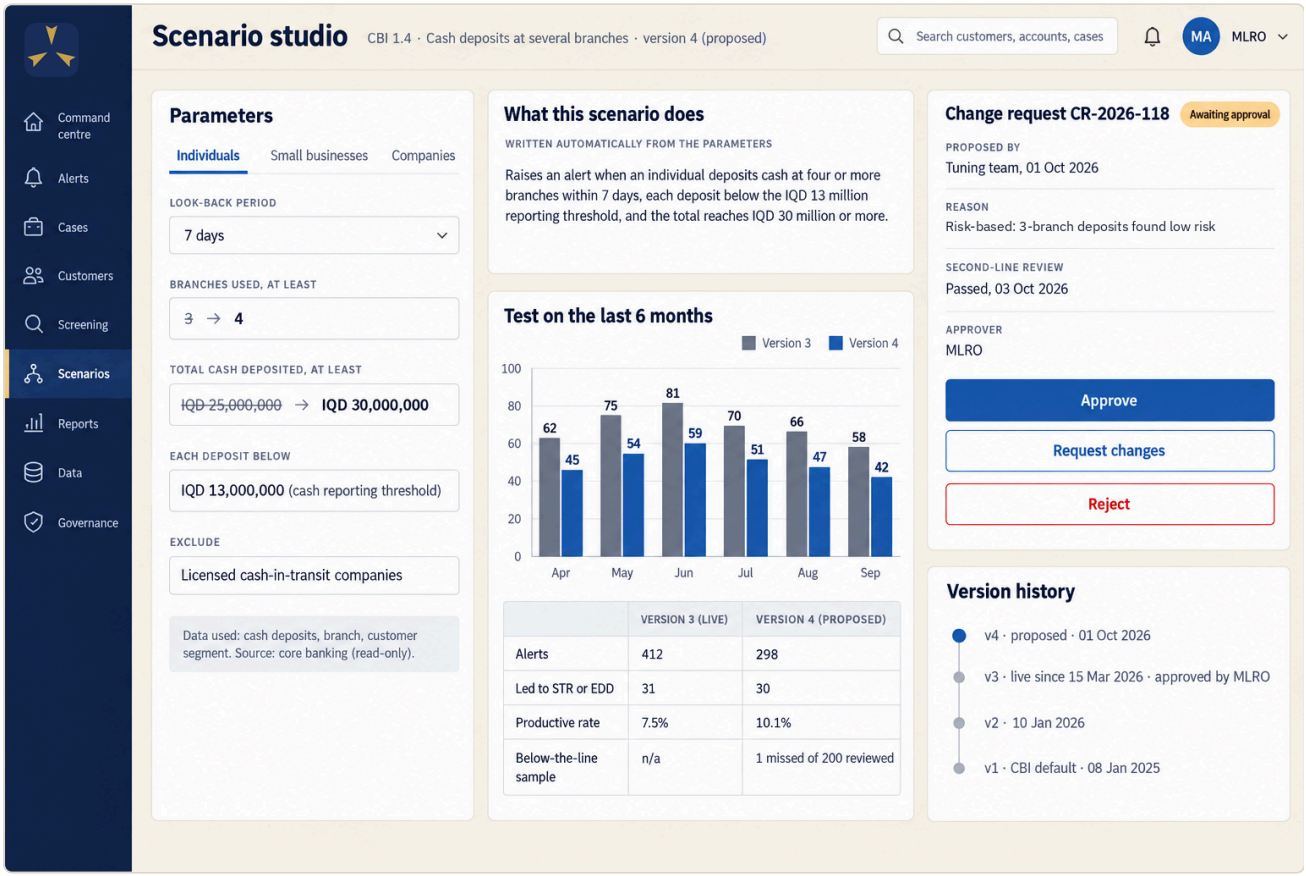


Figure 4. Scenario studio. A proposed change to CBI scenario 1.4 tested on six months of data: alerts fall from 412 to 298, and those that led to a report or enhanced due diligence from 31 to 30. The reason is recorded as risk-based, not workload, and the change now waits for the MLRO's approval.

Investigation and network analysis

An analyst opens a case, not a pile of alerts. The case shows why it was raised, how money moved in and out against what the customer declared, who the counterparties are, and what earlier alerts and reviews found.

- An AI assistant drafts a summary from the case data, labelled as a draft and linked to the transactions it relies on. The analyst edits it and decides.
- The network view links accounts through transfers, shared phone numbers, shared addresses and family records. That is how mule rings and hawala networks give themselves away.

- Closures and escalations follow a four-eyes rule, and every action is time-stamped in the case log.
- Requests for information to branches are tracked against deadlines, without alerting the customer.

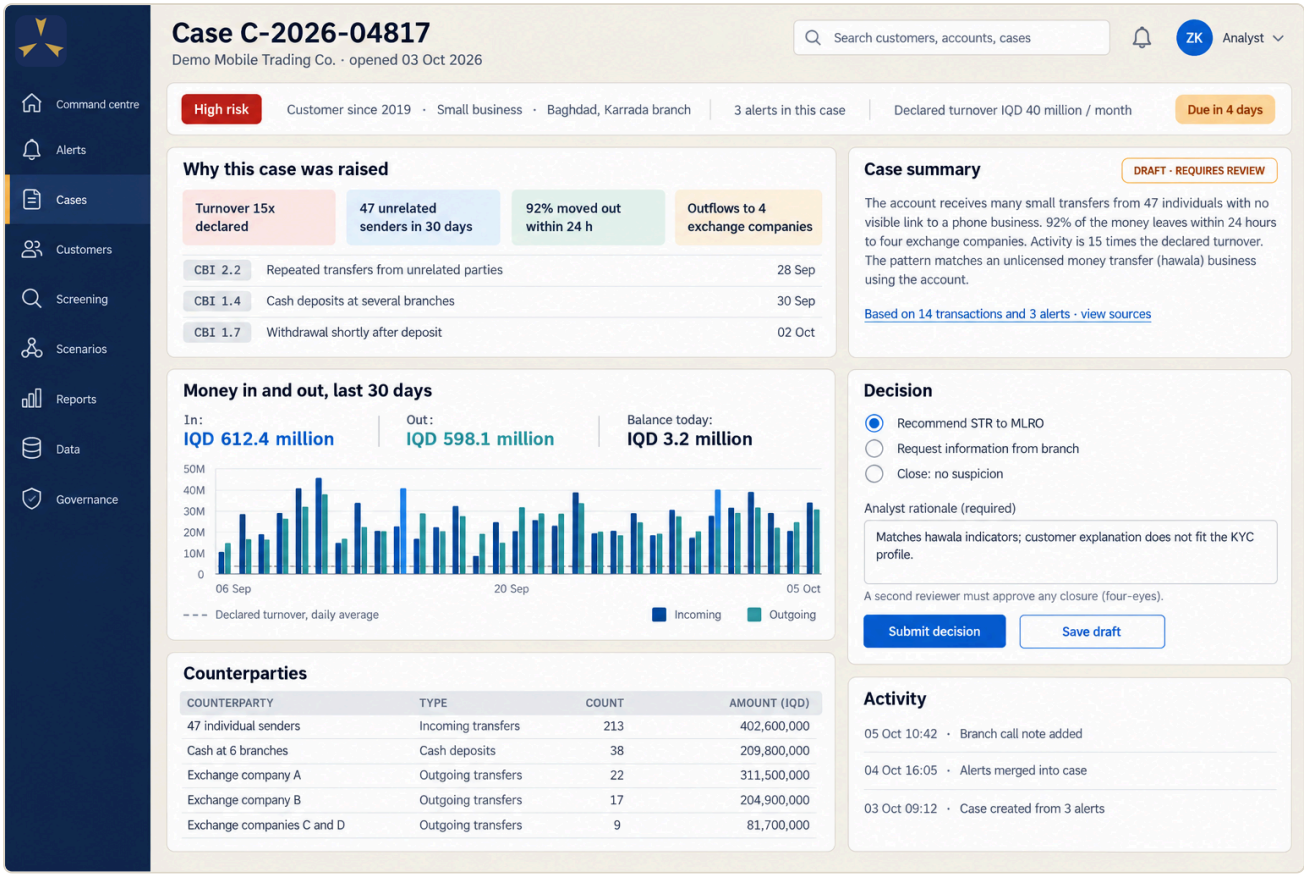


Figure 5. Case workspace. Three alerts on one customer grouped into one case: reasons, money in and out against declared turnover, counterparties, a draft summary awaiting review, and the four-eyes decision.

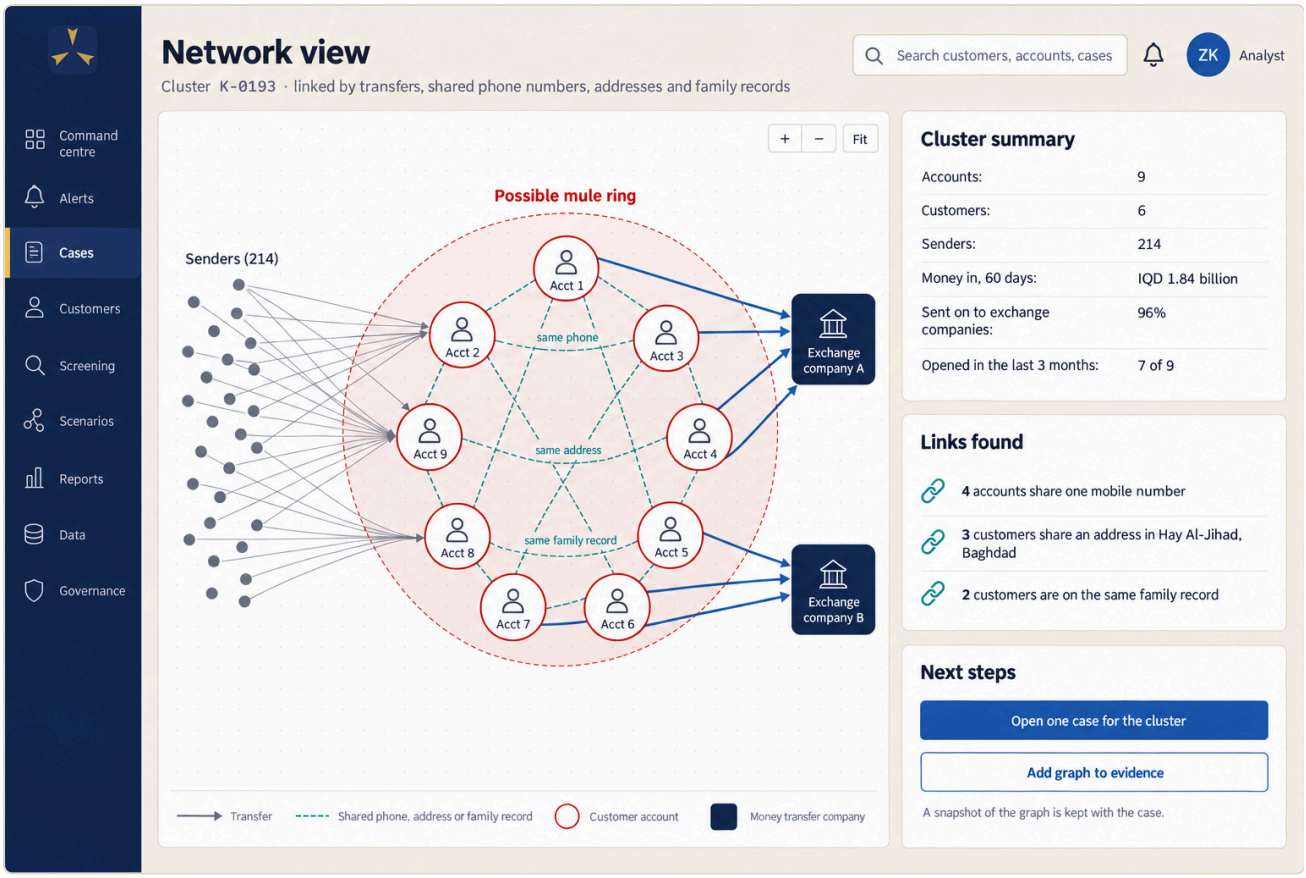


Figure 6. Network view. Nine accounts linked by transfers, a shared phone number, a shared address and a family record, all forwarding money to two exchange companies: one case for the whole cluster.

Reporting to the AML/CFT Office

Reports go to the AML/CFT Office through goAML, the reporting system the Iraqi financial intelligence unit uses. AKR-TMS builds each report from its case: subjects, accounts and transactions are filled from the core banking data, the narrative is drafted for the analyst to edit, and the report is checked against the goAML schema the Office publishes, including Iraqi rules such as the mother's name of each person reported, before the MLRO signs it. A named officer uploads the approved file and AKR-TMS records the Office's reference; it never holds a goAML password.

- STRs cover attempted as well as completed transactions, with indicators linked to the CBI scenarios.
- Unusual transaction reports, which the Office introduced in 2026 for activity that is unusual but not yet suspicious, are a third outcome besides an STR and no report. The customer is then watched more closely, and the case can still become an STR.
- Each draft is checked against the criteria the Office uses to grade every report it receives; it returns poor reports to the bank.
- CTRs and the other periodic reports are produced on schedule.
- The MLRO's written decision is recorded either way, including a decision not to report.
- Follow-up requests from the AML/CFT Office are tracked against the original report.

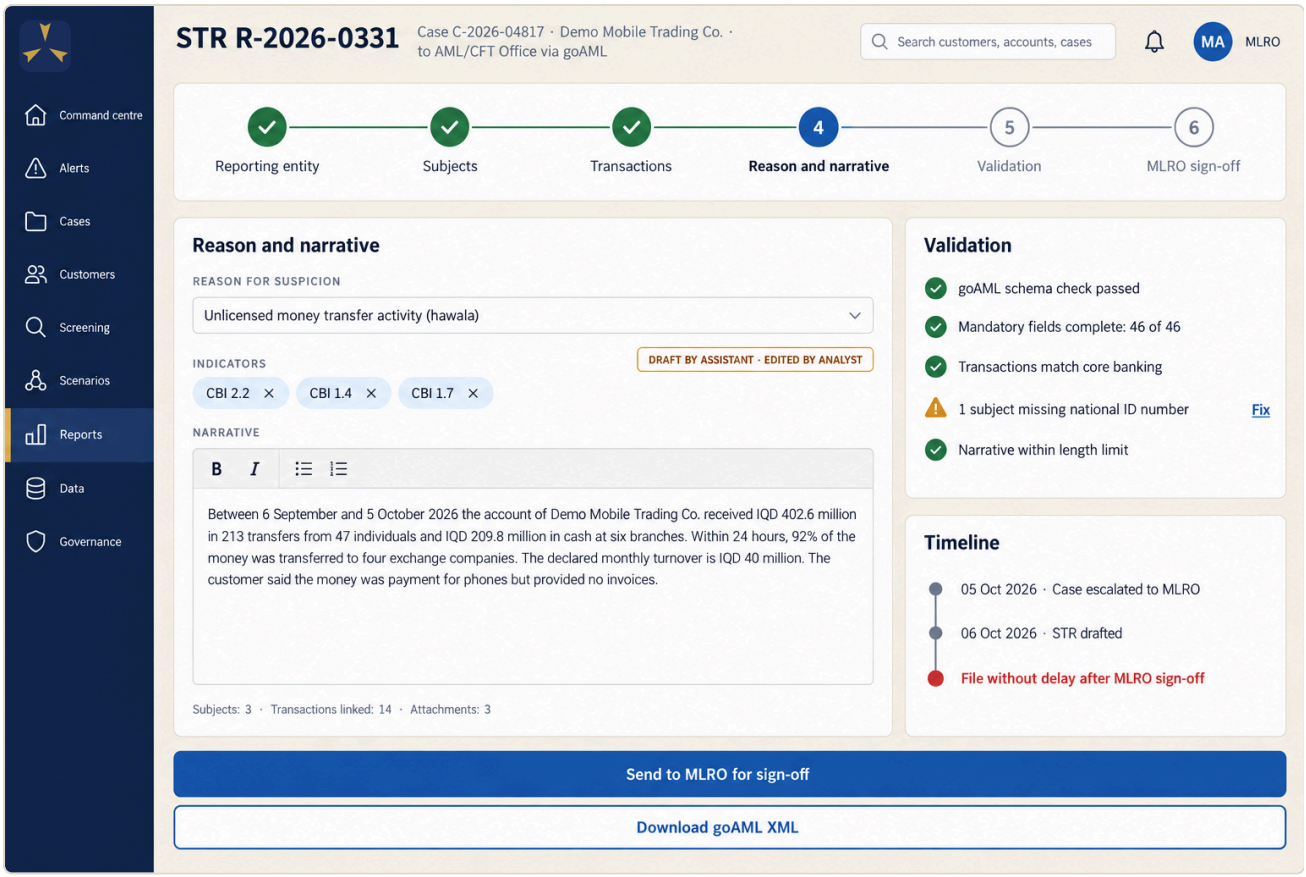


Figure 7. Report builder. An STR assembled from the case, validated against goAML's rules and waiting for the MLRO's sign-off. The narrative was drafted by the assistant and edited by the analyst.

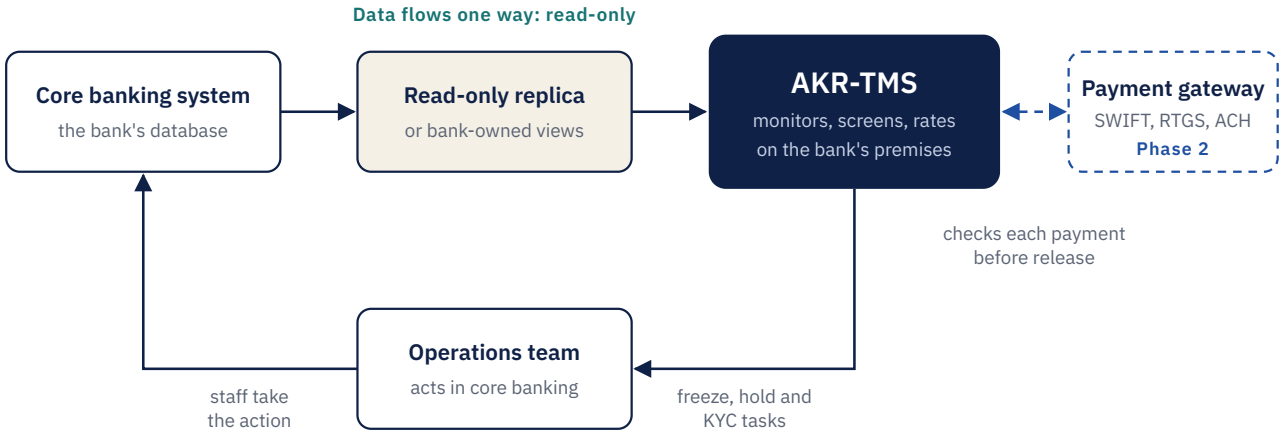
Governance and evidence packs

Everything that changes how AKR-TMS decides (a threshold, a scenario, a list source, a model) has an owner, a version, test results and an approval. The same records feed four evidence packs, one per audience:

AUDIENCE	WHAT ITS PACK SHOWS
CBI and AML/CFT Office	Coverage of the CBI catalogue, list freshness, backlog and timeliness, reports filed, branch liaison reports
Correspondent banks	Lists screened and update times, screening test results, scenario coverage, tuning log, backlog age, answers to their due diligence questionnaires
Independent validator, CBI-approved firm	Inventory of rules and models, test results, change history, data reconciliation, model documentation
Board and audit committee	Risk distribution and trends, overdue items, key changes and approvals, the annual certification

8. How it connects to the bank

AKR-TMS reads the core banking database and nothing else. It never writes to it and never calls the core banking application or its interfaces. That one rule makes AKR-TMS easy for a bank's IT and security teams to approve, and independent of which core banking system the bank runs.



How AKR-TMS connects. Data flows one way, from the core banking database to AKR-TMS. Actions such as freezes go to the bank's operations team as tasks; AKR-TMS then confirms from the replica that they were carried out. In phase 2 the payment gateway can ask AKR-TMS about each payment before releasing it.

What “read-only” means

- **It never changes data.** The database account AKR-TMS uses can only read. The bank’s own database enforces this, and the bank’s own audit log records every query.
- **It adds no load where it matters.** AKR-TMS reads from a replica, or from views the bank controls. Where no replica exists, it works within a load budget the bank signs off, and the bank can switch the connection off at any moment.
- **It sees everything.** Every day AKR-TMS reconciles what it received against the core banking system: record counts and totals per table, tied back to the general ledger. Any gap raises an alert, and each day’s result is signed and kept as a coverage certificate.

One product, any core banking system

Iraqi banks run several core banking systems: international products, regional ones and systems built in house. Some older systems cannot be read through a database connection at all. AKR-TMS translates each bank’s tables into one standard model of customers, accounts and transactions through a mapping layer. The mapping is configuration, owned by the bank and versioned like any other change, so connecting a new bank takes weeks of mapping rather than months of custom development. Where a database connection is impossible, the bank delivers files to an agreed landing area instead, with counts and checksums that still prove completeness. Payment messages are read from the payment systems’ own records, because a core banking system often holds only a summary of them.

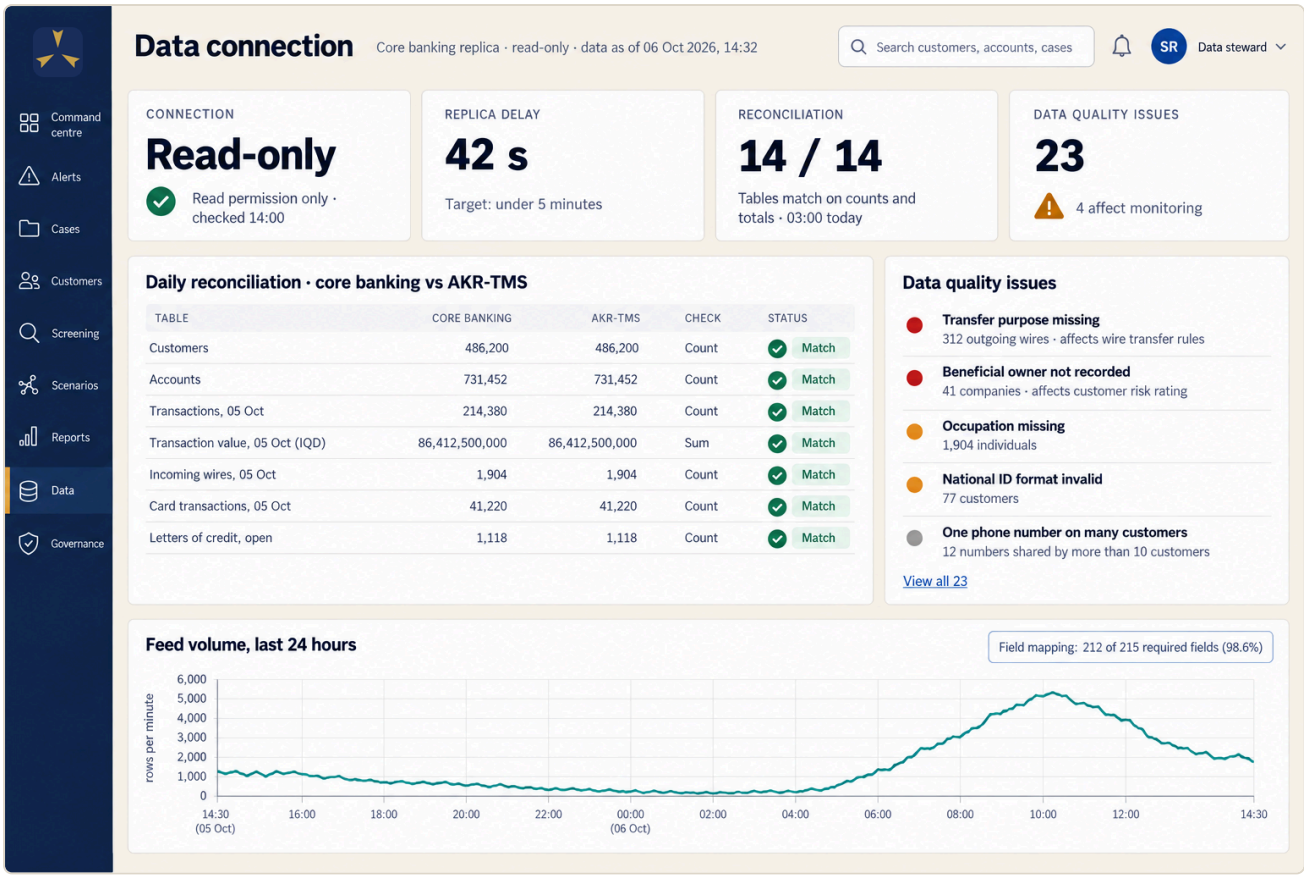


Figure 8. Data connection. The read-only status, replica delay, daily reconciliation of every table against the core banking system, and the data-quality issues that affect monitoring.

Acting without writing

When a customer must be frozen or a payment held, AKR-TMS does not do it itself. It creates a task for the bank's operations team, who act in the core banking system as they do today. AKR-TMS then checks the core banking data to confirm the action was taken, and keeps watching: any debit from a frozen account after the freeze, including bank fees, is flagged at once.

Phase 2: checking payments before they leave

The CBI requires a transaction that matches a scenario to be held until the MLRO clears it. For payments, that can only happen before release, inside the payment flow. AKR-TMS will offer an optional connection that the bank's payment gateway calls before it releases a SWIFT, RTGS or ACH payment. AKR-TMS screens and scores the payment and answers release, hold or reject within the time the payment system allows; the gateway then acts on the answer. The bank decides, rail by rail, what happens if no answer comes in time. The connection starts in shadow mode, then advises, and only then enforces. Until it is in place, AKR-TMS monitors minutes after the event, and records that real-time holds depend on the bank's existing payment filter.

Security

- Installed in the bank's own data centre; customer data never leave the bank.
- Encrypted connections, secrets kept in a vault, accounts with the least privilege they need, a separate network zone and no outbound internet access by default; list updates arrive through a

controlled channel.

- No on-demand look-ups of individual customers in the core banking database, so its logs cannot reveal who is under investigation.
- Evidence kept in write-once storage; every user action logged.
- Access by role: branch staff see the reasons for an alert, not the thresholds that produced it.

Under the hood. AKR-TMS uses a modern open-source stack: services written in Rust, PostgreSQL databases, write-once document storage, a secrets vault, single sign-on and a web interface built in React, running on a small Kubernetes cluster with full monitoring and logging. No part of it needs a foreign vendor's licence, and it runs without internet access.

9. AI, used responsibly

The CBI requires AML systems to use artificial intelligence and machine learning. Correspondents and the new EU rules require that people, not machines, make the decisions, and that every automated step can be explained. AKR-TMS meets both by putting each technique on one of three tracks, each with its own controls.

TRACK	WHAT IT COVERS	HOW IT IS CONTROLLED
Rules	CBI scenarios, the bank's own scenarios, screening settings	Readable in plain language; tested on past data before approval; versioned and reversible. Rules are configuration, not models
Models	Anomaly detection, alert prioritisation, network risk, calibration of risk scores	Documented; validated before use by someone independent of the builders; monitored monthly for drift; explained alert by alert; can be suspended at once, with automatic fallback to rules
AI assistant	Case summaries, report narrative drafts, Arabic-English translation, search across case documents	Drafts only, always labelled; every statement linked to its source; a named person edits and decides; runs inside the bank; never decides an alert or a report

- **Starter models, then the bank's own.** A new bank has few confirmed cases to learn from. AKR-TMS starts with models calibrated elsewhere, labelled as temporary, and builds the bank's own models from its decisions from the first day, under a recalibration plan with a target date.
- **Fair by design.** Iraqi names can reveal tribe, sect and ethnicity; these never enter a model. Nationality and geography are used only as documented risk factors that never trigger action on their own, and outcomes are compared across groups every month.
- **Built against automation bias.** The interface shows reasons before scores, requires a written rationale, and tracks how often analysts override the system. An unusual override rate triggers a review of the model.
- **Built to the strictest standard in the region.** The CBI has not yet issued rules on AI or model risk. AKR-TMS follows the UAE Central Bank's Model Management Standards, which cover AML models explicitly, together with UK, Singapore and US guidance, so it is ready for whatever the CBI-approved assessors and the correspondents test.

10. One product for many Iraqi banks

Iraqi banks differ widely: state banks with millions of customers and large branch networks, private banks of every size, Islamic banks and Islamic windows, banks headquartered in the Kurdistan Region, and branches of foreign banks. AKR-TMS is one product, configured for each bank and never customised.

Islamic banks

Murabaha, ijara, musharaka, mudaraba, salam, istisna'a and qard hasan each carry their own laundering risks: commodity trades that never take place, sale-and-buyback structures, investment accounts used to place cash, charity and zakat funds diverted to terrorism. AKR-TMS reads Islamic contracts as contracts, not as loans, and adds a scenario pack for them. Of the 28 banks on the CBI's February 2026 list of banks barred from dollar dealings, 17 are Islamic banks.

State banks

Large customer bases, wide branch networks and older systems. The file landing route connects systems that cannot be read through a database, and the branch window supports a liaison officer in every branch.

Kurdistan Region

Kurdish names are matched alongside Arabic ones. A Kurdish (Sorani) interface is planned next to Arabic and English.

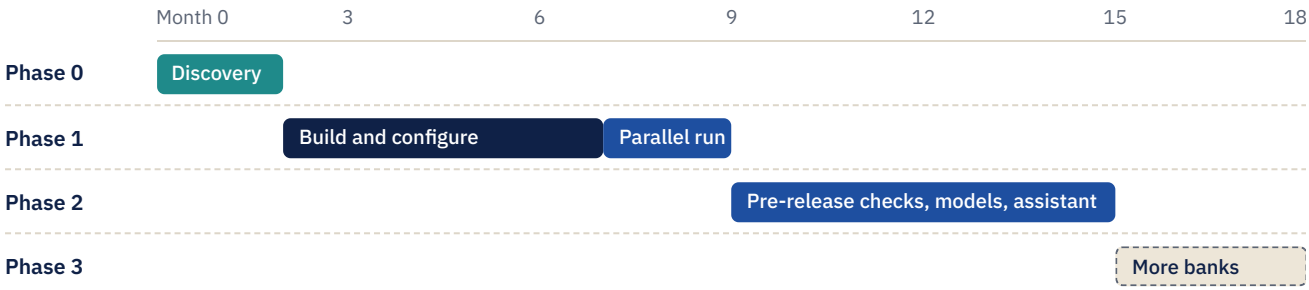
Regime packs

Iraq is always on. Packs for the US, the EU, the UK, the UAE, Saudi Arabia and Jordan add each regime's lists, rules, retention periods and evidence formats. A bank switches on the packs its correspondents and owners require. Where two rules differ, the stricter one applies unless the bank's policy records a reason otherwise.

Each bank gets its own installation in its own data centre, and no data are shared between banks. Product updates, list feeds and new scenarios arrive as signed packages, which the bank tests and approves before use.

11. Roadmap

AKR-TMS is delivered in phases, and each phase leaves the bank better off on its own. Durations are indicative for the first bank; they depend most on how fast database access is approved.



PHASE	WHAT IT DELIVERS
0. Discovery (6 to 8 weeks)	Inventory of the core banking and payment systems; read-only access approved; data mapped; data-quality baseline; alignment with the bank's own risk assessment
1. Core system (first bank live in about 9 months)	Screening with Arabic matching; the CBI scenario catalogue and the Iraq risk pack; customer risk rating and KYC reviews; cases, investigation and network view; goAML reporting; command centre, branch window and evidence vault; daily reconciliation. Ends with two to three months of running beside the bank's current system, comparing every alert
2. Before release, and smarter (about 6 months more)	Pre-release checks for payments; models in shadow mode, then live after validation; the AI assistant for summaries and narratives; correspondent and validator packs
3. Scale (ongoing)	Further banks in three to five months each; more regime packs; Islamic pack updates; Kurdish interface

12. What success looks like

100%
of core banking tables reconciled every day, and every gap explained

1 hour
at most from a list update to a full re-screen, against the CBI's 12-hour limit

0
high-risk customer reviews overdue

0
alerts older than 30 days without a recorded reason

Rising
share of alerts that lead to a report or enhanced due diligence

Minutes
to produce an evidence pack for the CBI or a correspondent

First time
pass of the CBI-approved firm's verification of the AML system

Weeks
of mapping, not months of development, to connect a new bank

AKR-TMS sets no target for the number of reports. Better detection will produce more reports, and Iraq's action plan asks for them, but a report filed to meet a quota is only noise for the AML/CFT Office.

Appendix A. Glossary

TERM	MEANING	ARABIC
AML/CFT	Anti-money laundering and countering the financing of terrorism	مكافحة غسل الأموال وتمويل الإرهاب
KYC, CDD	Know your customer; customer due diligence: identifying customers and understanding their activity	اعرف عميلك، العناية الواجبة تجاه العملاء
EDD	Enhanced due diligence: extra checks for higher-risk customers	العناية الواجبة المعززة
Beneficial owner	The person who ultimately owns or controls a customer	المستفيد الحقيقي
PEP	Politically exposed person: someone with a prominent public function, and their family and close associates	الشخص المعرض سياسياً
Risk-based approach	More effort where the risk is higher, less where it is lower	النهج القائم على المخاطر
Transaction monitoring	Checking account activity for signs of laundering or terrorist financing	مراقبة العمليات
Scenario	A rule that describes a suspicious pattern and raises an alert when it occurs	سيناريو
Alert, case	An alert is one signal; a case groups the alerts about one customer for investigation	تنبيه، حالة
False positive	An alert or list match that turns out not to be suspicious	إنذار كاذب
Screening	Comparing customers and payments against sanctions and PEP lists	الفحص على القوائم
Asset freeze	Blocking the funds of a listed person without delay	تجميد الأموال
STR	Suspicious transaction report to the financial intelligence unit	تقرير عن معاملة مشبوهة
CTR	Report of cash transactions above a threshold (USD 10,000 or its equivalent in Iraq)	تقرير المعاملات النقدية
goAML	Reporting software from the UN Office on Drugs and Crime, used by the Iraqi financial intelligence unit	goAML
AML/CFT Office	Iraq's financial intelligence unit, which receives STRs and CTRs	مكتب مكافحة غسل الأموال وتمويل الإرهاب
MLRO	Money laundering reporting officer: the bank officer who decides what is reported	مسؤول الإبلاغ عن غسل الأموال
Tipping off	Telling a customer that they are suspected or reported; prohibited	تنبيه العميل
Hawala	Informal money transfer outside the licensed system	الحوالة غير الرسمية
Structuring	Splitting transactions to stay below reporting thresholds	تجزئة المعاملات
Money mule	A person whose account moves other people's illicit money	وسيط نقل الأموال
Correspondent bank	A foreign bank that holds accounts for an Iraqi bank and makes payments on its behalf	المصرف المراسل
Grey list	The FATF's list of jurisdictions under increased monitoring	القائمة الرمادية
Mutual evaluation	The peer review of a country's AML/CFT system by the FATF or a regional body	التقييم المتبادل
Core banking system	The bank's main system of record for customers, accounts and transactions	النظام المصرفي الشامل
Three lines of defence	Business units, compliance and risk, and internal audit, each with its own control role	خطوط الدفاع الثلاثة
Regime pack	An AKR-TMS component that adds one foreign regime's lists, rules and record-keeping	حزمة الأنظمة الأجنبية

Appendix B. The research behind this plan

The plan rests on the research files listed below, written in October 2026 and delivered with it. Each file records its sources and ends with requirements for AKR-TMS, each traced to a law, a standard, a regulator's finding or a market fact.

RULES AND STANDARDS

01 Iraq: AML/CFT law and CBI rules

02 Iraq: money laundering and terrorist financing risks

03 FATF standards

04 Basel, Wolfsberg, Egmont and UN sanctions

05 US, EU, UK and Gulf regimes

06 AML and KYC fundamentals, with an English-Arabic glossary

THE MARKET

07 Vendor landscape overview

08 NICE Actimize

09 Oracle FCCM

10 SAS

11 SymphonyAI, FICO Siron, Nasdaq Verafin, Fiserv

12 AI detection specialists

13 Investigation automation and challengers

14 Screening engines

15 Risk data providers

16 KYC and client lifecycle platforms

17 AML modules of core banking systems

18 The Iraqi and regional market

LESSONS AND DESIGN

20 Enforcement failures

21 Detection technology

22 Arabic, Kurdish and Iraqi name matching

23 Regulatory reporting and goAML

24 Islamic banking

25 Core banking integration

26 Customer risk rating and due diligence

27 Monitoring scenarios

28 Case management and workflow

29 Governance and model risk

30 Security and privacy

31 Analyst experience

32 Technology stack

RESEARCH AREA	FILES	REQUIREMENTS
International standards: FATF, Basel, Wolfsberg, Egmont, UN sanctions	03, 04	78
Foreign regimes: US, EU, UK, Gulf	05	61
The market	07-18	249
Enforcement	20	68
Detection technology	21	66
Arabic, Kurdish and Iraqi name matching	22	59
Regulatory reporting and goAML	23	84
Islamic banking	24	30
Core banking integration	25	65
Customer risk rating and due diligence	26	42
Monitoring scenarios	27	60
Case management and workflow	28	76
Governance and model risk	29	54
Security and privacy	30	102
Analyst experience	31	71
Technology stack	32	43
Total		**1,208**

Iraqi law and the CBI's instructions (file 01) are the baseline for all of them; their requirements are summarised in section 3.